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Federated Learning for Intelligent Transportation Systems: Use Cases, Open Challenges, and Opportunities

Abstract—Intelligent transportation systems (ITSs) leverage a network of interconnected infrastructures utilizing advanced technologies to improve traffic management and safety. Federated learning (FL) has emerged as a pivotal method within ITSs, enabling decentralized collaborative model training without direct data sharing, thus preserving privacy and enhancing system efficiency. This article explores the integration of FL in ITSs, focusing on FL's application in traffic flow prediction, trajectory prediction, parking space estimation, and traffic target recognition. Despite its potential, FL deployment faces challenges, including data heterogeneity, communication and bandwidth constraints, and resource limitations on edge devices. Addressing these challenges is crucial for realizing the full potential of FL in ITSs. This article provides a comprehensive survey of existing FL implementations in ITSs, discusses inherent challenges, and outlines future research directions aimed at overcoming these obstacles.

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Introduction

Intelligent transportation systems (ITSs) refer to networks of interconnected transportation infrastructures that mutually communicate using sophisticated technologies. ITSs are designed to provide innovative services related to various transport modes and traffic management, thus empowering users to be better informed and to make safer, more coordinated, and smarter utilization of transport networks. ITSs integrate cutting-edge sensing technologies, such as traffic cameras, to compile data on traffic conditions, vehicle counts, and environmental factors. These systems also feature advanced software systems that analyze sensory data to monitor traffic flows and administrate traffic signals accordingly. Communication networks within ITSs facilitate data transfer among various components of the system, encompassing vehicle-to-vehicle (V2V) and vehicle-to-infrastructure communications. In addition, control systems within ITSs automate procedures, such as dynamic traffic routing.

The components of ITSs include vehicle nodes, sensors, roadside units (RSUs), base stations, and other communication technologies. These components work together to facilitate various functions, such as traffic management, route planning, and safety enhancements. ITSs leverage a variety of sensors and data sources to collect relevant information for traffic management and decision making. Vehicles themselves act as moving sensors, equipped with onboard units that serve as vehicle nodes. Fixed RSUs act as the primary steadfast nodes, with dual communication interfaces to interact with stationary and mobile entities. To integrate data from various sources effectively, data fusion techniques are often utilized. Through the connectivity offered by ITSs, vehicle nodes can access a multitude of services, including infotainment, emergency support, and traffic management [1].

The deployment of ITSs can improve the utilization of transportation resources and improve traffic efficiency by leveraging machine learning (ML) and deep learning (DL) techniques, such as reinforcement learning (RL) [4] and federated learning (FL) [7]. RL enables an agent to learn by interacting with its environment. By performing actions and receiving feedback in the form of rewards or penalties, the agent refines its decision-making process to maximize long-term rewards. This characteristic makes RL particularly well suited for ITS applications, as it allows the system to adapt to the dynamic and often unpredictable nature of traffic conditions [20], [21]. FL is a distributed ML technique where geographically distributed devices, such as vehicle nodes and RSUs in an ITS network, can cooperatively train a shared predictive model without directly exchanging raw data. This ensures data privacy by keeping training data localized on individual devices. The collaborative model training process relies on the exchange of model updates, not the underlying data themselves. Each device trains the model locally using its data and transmits the resulting updates to a central server for aggregation, ultimately refining the global model. This decentralized method offers significant advantages in ITSs, enabling applications like traffic flow optimization, real-time decision support for autonomous vehicles, and enhanced route planning, all while mitigating privacy concerns [18].

Despite the advancements and potential benefits of vehicular edge computing and FL, many existing applications still predominantly rely on centralized training models. Such deployment raises several concerns [8]:

- **Privacy and data security:** Centralized models require the transfer of large volumes of data from local environments to central servers. This aggregation of sensitive data can increase the risk of privacy breaches and

cyberattacks, making the system vulnerable to unauthorized access and data theft [3], [5].

- *Poor real-time performance:* Decisions often need to be made in real time to effectively manage traffic and ensure safety. Centralized processing can introduce unacceptable delays as data travel to and from the central server, potentially leading to slower response times during critical moments [24].
- *Bandwidth consumption:* Transmitting large datasets over the network to a central server consumes substantial bandwidth. In densely populated urban areas or during peak times, this can lead to network congestion, further exacerbating latency issues and reducing the overall system efficiency.
- *Scalability limitations:* As the number of connected devices in ITSs increases, centralized systems may struggle to scale efficiently. Centralized systems may need to be upgraded with increased data volume, which may be costly.
- *Single point of failure:* Centralizing the training and data processing creates a single point of failure, making the entire system more susceptible to outages and attacks. If the central server fails, the entire ITS could be compromised, leading to widespread disruptions [23].
- *Data management challenges:* Centralized systems must handle potentially heterogeneous data from diverse sources, which can complicate data management, integration, and analysis efforts.
- *Regulatory and compliance issues:* With stringent regulations, like the General Data Protection Regulation in Europe and various local data protection laws, managing compliance can be more challenging with centralized systems, due to the diverse nature of the data and their widespread collection and processing locations [22].

These concerns have prompted the development of decentralized training methodologies aimed at addressing the challenges and enhancing the efficiency and security of data processing in ITSs. The utilization of vehicle nodes is an innovative strategy for enhancing the execution of FL tasks, allowing the offloading of computational processes from smart vehicles to the edge. Vehicular edge computing offers a solution by optimizing computational efficiency through the distributed processing capabilities of edge devices. This facilitates the effective utilization of resources, enabling real-time analysis and processing of data at the network edge. Such methods provide immediate actionable insights based on prevailing road conditions and available resources, significantly enhancing the responsiveness of ITSs [9].

Although FL is well suited for decentralized training methods, its deployment within the complex and dynamic environment of ITSs encounters significant challenges, such as stringent quality-of-service demands and the need for dynamic resource allocation. This article addresses

these challenges by conducting a comprehensive survey of existing solutions that integrate FL with ITSs. It also outlines technical challenges and discusses directions for future research.

FL

FL represents an innovative method to distributed model training. It serves as a crucial enabling technology for various emerging applications, including ITSs. FL in ITSs relies on several essential building blocks:

- *FL clients:* These edge devices or local systems are physically distributed and can include RSUs and vehicular nodes.
- *Central server:* The central server coordinates clients' local models and manages the aggregation of local model updates received from the clients. This server is responsible for generating an initial global model, which is disseminated to all FL clients. It also updates the global model as new trained local models are received from the clients.
- *Global model:* This ML model is shared across all participating clients. It serves as the starting point for local training on clients and is incrementally improved through learning from the collective updates received from all participating clients.
- *Local model:* Each client trains a local model based on its own data, starting from the global model parameters received from the central server.
- *Communication protocol:* The method and frequency of communication between the central server and the clients are crucial for efficiency and effectiveness. This includes how often local model updates are sent to the central server, which can significantly impact the speed of learning and network load.
- *Aggregation algorithm:* This algorithm is used by the central server to combine updates from various clients into a single improved global model.

The process begins with the FL server generating a global model, which is then disseminated to all FL clients. The global model represents the starting point for collaborative learning and is transmitted securely to ensure its integrity and confidentiality. Upon receiving the model, each client utilizes its locally stored data for individual training. Local training allows the model to capture insights specific to local traffic conditions, road infrastructure, and environmental factors. During this phase, the federated averaging algorithm coordinates the aggregation of local model updates while respecting data locality. The locally stored data never leave the client, to preserve privacy. Since clients collect and store distinct datasets, the training data vary across clients.

Once local training is completed, a secure aggregation mechanism facilitates the combination of local model updates in a privacy-preserving manner. This aggregated

update is then transmitted to the central server, allowing for the refinement of the global model. The aggregated model updates are integrated into the global model, leading to its refinement based on the collective knowledge learned from diverse local datasets. This iterative process of model refinement enhances the global model's accuracy in predicting traffic flow, recognizing traffic targets, and optimizing vehicular edge computing. The refined global model is then re-deployed to the client nodes, enabling them to make informed traffic management decisions based on the updated model. The local deployment of the refined model supports real-time inference and decision making, contributing to the overall efficiency of ITSs. Figure 1 illustrates the high-level process of FL.

The frequency and method of communication between the central server and the clients are crucial for ensuring efficiency and effectiveness. FL enhances data isolation during training, safeguards data privacy, and facilitates efficient collaborative learning among multiple vehicular edge devices within ITSs. By enabling both data storage and model training to occur locally, FL fosters the collaborative development of globally optimal models. Moreover, by relying solely on interactions with a central server for model updates, FL shows significant potential to improve the efficiency and security of ML in ITS applications [6].

FL in ITSs: Use Cases

The distributed learning paradigm of FL allows vehicles and RSUs to collaboratively improve models without directly sharing raw data, enhancing operational efficiency, safety, and data privacy in ITSs. This section reviews use cases of FL in ITSs, synthesizing findings from recent research and identifying trends and gaps in the literature.

Traffic Flow Prediction and Congestion Management

ITSs play a crucial role in traffic flow prediction by leveraging a diverse array of data sources, as demonstrated in Figure 2. Data are analyzed to estimate future traffic conditions, which helps to manage road resources, alleviate congestion, and improve traffic efficiency. These sources include mapping data, historical and real-time traffic information, and external factors, such as weather conditions. By analyzing

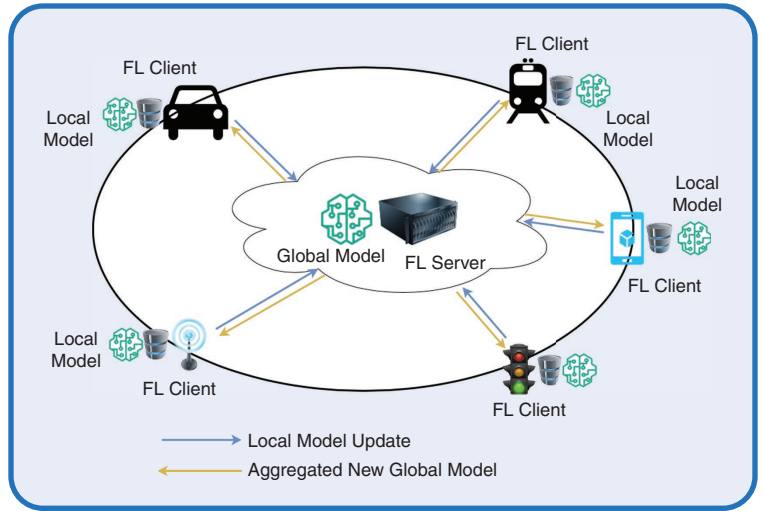


FIG 1 The high-level process of FL.

this comprehensive dataset, an ITS can estimate future traffic conditions, enabling more effective road resource management, congestion alleviation, and overall improvement in traffic efficiency.

ML and DL techniques have emerged as powerful tools in this domain, capable of extracting and learning complex traffic flow features from historical data. The accurate predictions generated by these models facilitate the implementation of strategic interventions, such as optimal route recommendations and adaptive road signal timing. These measures contribute to smoother traffic flow and reduced congestion across transportation networks. Furthermore, the ability to predict traffic flow allows for timely detection and notification of incidents like accidents or congested

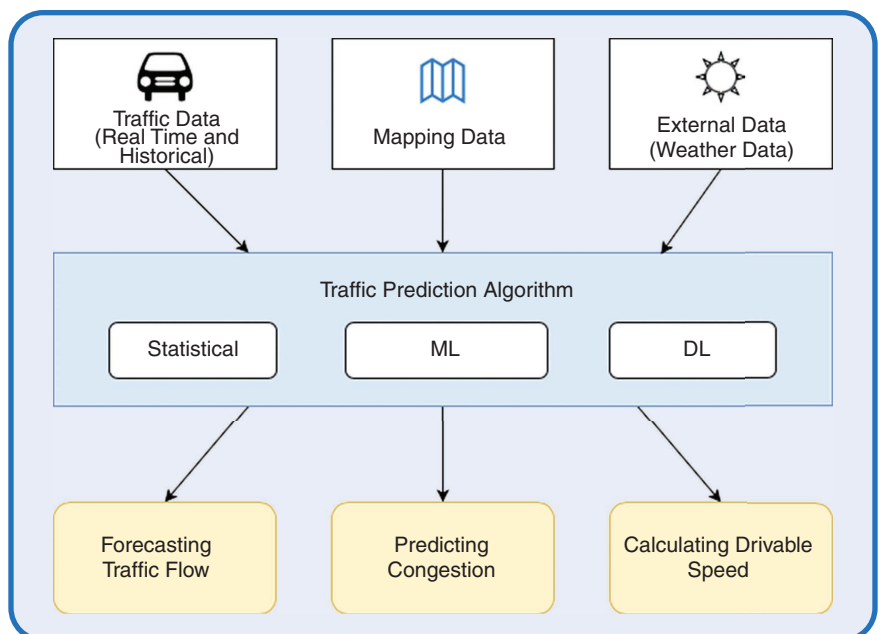


FIG 2 The traffic flow prediction and congestion management in ITSs.

roads. This early warning system empowers authorities to respond quickly and proactively to potential problems, improving overall road safety and efficiency [25], [26], [51].

The effectiveness of these systems faces challenges, particularly in the form of spatiotemporal data gaps within traffic acquisition systems. These gaps can significantly hinder the accuracy of traffic state estimation, potentially impacting the reliability of predictions and subsequent decision-making processes [52]. Addressing these data quality issues and privacy concerns is crucial for improving the robustness and reliability of traffic flow prediction in ITSs.

In response to these challenges, FL has emerged as a promising approach for training models that predict traffic flow patterns across cities while preserving data privacy. FL enables collaborative model training without sharing raw data, allowing multiple parties to contribute to a global model while keeping sensitive information secure. This approach not only addresses privacy concerns but also has the potential to mitigate data quality issues by leveraging diverse data sources and local expertise, ultimately enhancing the accuracy and reliability of traffic flow predictions in ITSs.

Liu et al. [27] address the challenge of performing accurate traffic flow prediction by proposing an FL-based gated recurrent unit (FedGRU) neural network algorithm. This method differs from centralized learning approaches by updating a universal learning model through a secure parameter aggregation mechanism. The proposed FedGRU algorithm is enhanced with a secure parameter aggregation mechanism using a federated averaging algorithm to reduce communication overhead. An ensemble clustering-based scheme is also included to improve traffic flow prediction by grouping organizations into clusters before applying the FedGRU algorithm. Extensive case studies show that FedGRU's prediction accuracy exceeds 90%, confirming its efficacy in providing accurate and timely traffic predictions without compromising the privacy and security of raw data. However, the potential for communication overhead during model updates remains a concern.

Liu et al. [48] introduce deep traffic state prediction (DeepTSP), a DL model addressing TSP challenges like scale, granularity, and sparsity. While outperforming traditional methods, DeepTSP faces challenges in scalability and data dependency, which are addressed more effectively in FL frameworks. Addressing these limitations, Sepasgozar et al. [28] present an FL-based method to predict network traffic flow in smart cities, with an emphasis on privacy preservation. The authors leverage long short-term memory (LSTM) to train localized models without transferring sensitive data, thus ensuring privacy. The main aims are to tackle the challenges of accuracy, data volume, and data privacy in network traffic prediction within ve-

hicular ad hoc networks (VANETs). The proposed solution is characterized by its decentralized ML structure, which outperforms other ML algorithms in terms of prediction errors and achieves a high R^2 score. This demonstrates its effectiveness in real-world network conditions while prioritizing data security. However, the method's extensive computational resource requirements for training remain a challenge.

Yuan et al. [29] further advanced the field with federated DL based on spatial-temporal long and short-term networks (FedSTN), a federated DL method that utilizes graph representations to mine spatiotemporal information from traffic data. FedSTN's architecture consists of modules for capturing long-term spatial-temporal dynamics, sharing short-term information while preserving privacy through homomorphic encryption and recognizing semantic features like non-Euclidean connections and points of interest. This method aims to ensure data privacy and efficient model training without centralizing data. Simulations conducted on practical datasets validate the effectiveness of the FedSTN algorithm, suggesting it as a promising solution for accurate traffic flow prediction. However, the model's complexity and the need for high computational resources may limit its scalability and practical deployment.

To address the high time cost associated with DL models, Qi et al. [31] propose a novel DL framework named FedAGCN, aimed at improving real-time traffic flow prediction. This improvement is achieved by integrating FL with asynchronous graph convolutional networks (GCNs). A graph FL strategy named GraphFed manages parameter updates across the federated network to maintain prediction accuracy. The framework has been tested on two public traffic datasets, and the results demonstrated that FedAGCN not only maintains high prediction accuracy but also significantly reduces both the training and inference times of the DL model. This method addresses the challenges faced by centralized models, which suffer from transmission delays due to the bandwidth required to transfer data from traffic nodes to a central server. The framework effectively navigates the issue of spatial correlation in traffic data, which can vary from one area to another. However, challenges remain in handling asynchronous updates and potential latency in federated settings.

Xia et al. [32] introduce a model that combines FL with a GCN to enhance the performance of short-term traffic flow prediction. The model incorporates community detection to identify local subnetworks within the traffic network, allowing for more relevant and localized training that leads to potentially more accurate predictions. GCNs are utilized to capture the spatiotemporal characteristics of traffic flow data, effectively leveraging the topological structure of the traffic network. This method aims to address common challenges in global GCNs, including high training times, communication costs, and data privacy

Table 1. A summary of FL applications for traffic flow prediction and congestion management in ITSs.

Reference	Methodology	Model	Outcomes	Challenges
[26]	FedGRU with secure parameter aggregation and federated averaging	FL, GRU, federated averaging	Exceeds 90% accuracy in traffic predictions and preserves data privacy	Potential communication overhead during model updates
[48]	DL model leveraging spatiotemporal data for TSP	DeepTSP	Significantly improves accuracy of TSPs	High computational requirements, scalability issues
[28]	FL-based method using LSTM	FL, LSTM	Achieves a high R^2 score and enhances accuracy and privacy in VANETs	Extensive computational resources required for training
[29]	FedSTN utilizing graph representations to mine spatiotemporal information from traffic data	Federated DL, graph representations	Enhances traffic management and driver guidance in smart cities	High computational requirements, scalability issues
[31]	FedAGCN framework integrates FL with asynchronous GCN	FL, asynchronous GCN	Reduces training and inference times and maintains high accuracy	Requires robust infrastructure for handling asynchronous updates, potential high latency
[32]	FL combined with GCN for short-term traffic flow prediction incorporating community detection	FL, GCN	Outperforms benchmark models and improves prediction accuracy and security	High computational requirements, scalability issues

risks. The proposed GCN outperforms several benchmark models, including LSTM, convolutional neural network (CNN), Chebyshev spectral CNN, and graph attention network models. The model predicts traffic states accurately and securely by aggregating local model parameters into a global model. While outperforming several benchmark models, this method faces potential scalability issues and high computational costs associated with GCNs.

Most recently, Xu et al. [47] presented the attentive graph neural process, combining Gaussian processes and neural networks to model traffic speed as a stochastic process. This method effectively handles missing data and provides a probabilistic framework for traffic speed prediction. However, its computational intensity and potential scalability issues in large-scale traffic networks remain areas for further research.

Table 1 summarizes FL applications for traffic flow prediction in ITSs.

Trajectory Prediction

Vehicle trajectory prediction is a critical component in various applications, such as autonomous driving, traffic management, collision avoidance, and route planning. It involves forecasting the future path or position of a vehicle over a certain period, based on the vehicle's historical and real-time location data, movement patterns, and surrounding traffic conditions, as shown in Figure 3.

Traditional trajectory prediction models typically use time-series data, including information on vehicle or passenger position, velocity, and acceleration. These models often harness the capabilities of deep neural networks, particularly recurrent neural networks (RNNs) and trans-

formers, which have shown effectiveness in forecasting trajectories of various entities, such as vehicles and pedestrians, and in capturing their behavioral patterns [33]. Chen et al. [49] highlight a crucial aspect of trajectory prediction: ensuring scene consistency. Their work emphasizes the need to prevent unrealistic predictions and potential self-collisions, which is particularly important in complex traffic scenarios. This research underscores the importance of considering the broader context and environment in which vehicles operate, rather than treating each trajectory in isolation. Although traditional centralized ML approaches have shown promise, they often require large amounts of sensitive vehicle data, raising privacy concerns. Addressing these concerns, FL has emerged as an alternative approach. FL allows multiple parties to collaboratively train a model without sharing raw data,

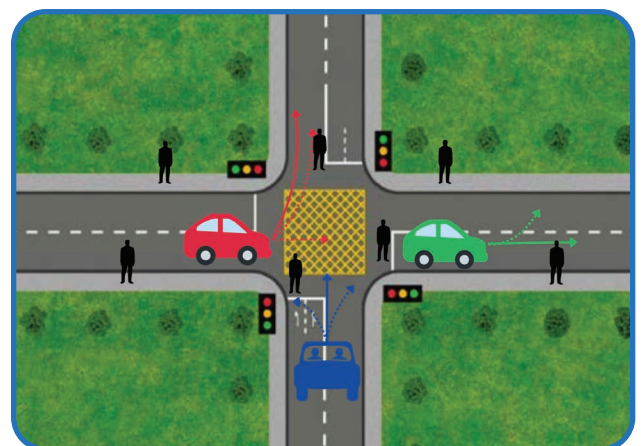


FIG 3 The trajectory prediction.

incorporating information from various sources, such as different vehicles, traffic cameras, or infrastructure sensors while maintaining privacy.

Wang et al. [2] propose a hierarchical trajectory anomaly detection scheme for ITSs utilizing hierarchical FL and blockchain technologies. This approach enhances the generalization capability of the global trajectory anomaly detection model by incorporating multiarea data through secondary fusion. The integration of blockchain and FL facilitates the iterative exchange and fusion of the global trajectory anomaly detection model through coordinated on-chain and off-chain data access. The scheme is important to understand the motion behavior of dynamically evolving vehicular edge devices, predict their future motion trends, and provide support services for ITSs.

Zhou et al. [34] combine FL with spatiotemporal analytics for vehicle trajectory prediction. The spatiotemporal FL (STFL) technique incorporates both the spatial dimensions (like location and geography) and temporal dimensions (such as time-series data) into an ML model, while FL clients contribute to the training of a shared model without needing to exchange raw data, thereby protecting privacy. Further advancing the field, Wang et al. [50] present a novel approach combining automated ML with FL to enhance trajectory prediction models. Experiments demonstrate that Automatic Trajectory Prediction Model Design under Federated Learning Framework (ATPFL) outperforms traditional models by providing higher accuracy and efficiency in predictions.

Koetsier et al. [35] apply a one-class support vector machine (OCSVM) and an isolation forest within an FL framework to detect anomalous behavior in vehicle trajectories. The authors utilize FL collaboratively to identify anomalous vehicle trajectories at multiple intersection settings, including DR USA Intersection EP (EP), DR USA Roundabout SR (SR), and DR USA Intersection MA (MA). The OCSVM works by learning a function that captures the majority of data points (the normal class) and identifies any points that do not fit this function as anomalies. It effectively creates a boundary around the normal data points, and anything that falls outside this boundary is considered

anomalous. The isolation forest builds an ensemble of isolation trees for the high-dimensional datasets. The average area under curve (AUC)–receiver operating characteristic curve score showed an improvement from 97% observed in models trained independently at each intersection to up to 99% when using the federated model.

Table 2 summarizes various FL methods for predicting vehicle trajectory.

Parking Space Estimation

Efficient parking space management is a critical component of smart city initiatives, aiming to reduce traffic congestion, minimize fuel consumption, and improve overall urban mobility. Accurate estimation of available parking spaces in real time has become increasingly important as cities grapple with growing populations and limited infrastructure. Traditionally, DL approaches have shown promise in tackling this challenge, leveraging data from various sources, such as surveillance cameras, Internet of Things (IoT) sensors, and historical parking patterns to predict parking availability.

Traditionally, DL models, particularly CNN [53], [54], [55] and LSTM [56], [57], [58] models, have shown promise in tackling this challenge. These models can process complex visual data from parking lot cameras, analyze temporal patterns in occupancy rates, and incorporate external factors like weather conditions or local events to improve predictions. However, the effectiveness of these centralized DL approaches often relies on aggregating large amounts of data from multiple parking facilities, raising significant privacy concerns and potential regulatory challenges.

FL has emerged as a promising solution to address these limitations while maintaining the benefits of DL for parking space estimation. FL enables collaborative model training across multiple parking facilities or data sources without centralizing raw data. Recent research has explored various applications of FL in the estimation of parking spaces, each addressing different aspects of the challenge.

Huang et al. [19] propose a system where multiple parking lot operators collaborate to train an LSTM model without sharing raw data, thereby preserving privacy. An

Table 2. Various FL methods for vehicle trajectory prediction.

Reference	Method	Key Features	Benefits
[33]	DL with RNNs and transformers	Utilizes time-series data (position, velocity, acceleration)	Forecasts future vehicle paths based on historical and real-time data
[2]	FL with blockchain	Hierarchical FL for the detection of anomalies in the trajectory	Blockchain integration for secure model exchange and fusion
[34]	FL with spatiotemporal analytics (STFL)	Combines FL with spatiotemporal data for trajectory prediction	FL clients train shared model without sharing raw data
[35]	FL with OCSVM and isolation forest	FL for collaborative identification of anomalous trajectories	Detects and analyzes anomalies in vehicle trajectories at various intersections

incentive mechanism using a Stackelberg game is proposed to recruit parked vehicles for edge computing tasks, optimizing resource utilization and enhancing system capacity. The proposed method offers several advantages, including improved data privacy, efficient utilization of idle vehicle resources, and enhanced computational capacity for real-time data processing. Numerical results demonstrate the effectiveness of federated parking in providing accurate parking space estimations and managing computational workloads through a distributed privacy-preserving approach. However, the research reveals several gaps, such as scalability and real-world implementation challenges, refinement of incentive mechanisms, and optimization of communication protocols to reduce latency and overhead associated with frequent model parameter updates.

Similarly, Yuan et al. [59] present a novel approach to collision-free trajectory planning for automated parking in narrow spaces using an FL scheme. The proposed method, hierarchical trajectory planning method with deep reinforcement learning in the federated learning scheme (HALEOS) integrates deep RL (DRL) with optimization-based techniques to generate efficient and accurate parking trajectories. The hierarchical approach improves planning time by 12.15%–66.02% compared to baseline methods while maintaining trajectory accuracy and enhancing generalization capabilities through decentralized training. The use of FL protects vehicle data privacy during model parameter aggregation. However, the FL process may experience instability in early training stages and requires substantial computational resources for model updates.

Senapati et al. [36] compare several DL models, including artificial neural network (ANN), CNN, and LSTM models, to determine the optimal model for parking space estimation in an FL context. Their study integrates factors such as holidays, weekends, and weekdays to enhance the dataset, with results showing that the LeNet-5 architecture performs best in a traditional setup, while the LSTM model achieves the highest accuracy in the FL setup. This study highlights the potential of FL in maintaining data privacy while achieving accurate parking space predictions. However, it also indicates challenges, such as communication overhead and the need for significant computational resources for model training and parameter updates.

Similarly, Sha et al. [60] present a novel methodology that combines FL with DL models to predict parking slot availability. This approach employs a dual strategy, utilizing both spatial and temporal data to train models separately before aggregating them into a unified predictive model. The spatial analysis examines the relationship between parking space occupancy and environmental characteristics, such as proximity to entrances and structural characteristics. Concurrently, the temporal analysis captures patterns related to the time of day, the day of the week, and seasonal variations. These insights are then combined using federat-

ed aggregation to create a comprehensive model that leverages both spatial and temporal data while ensuring privacy. The system integrates various DL architectures, including ResNet, DenseNet, EfficientNet, and VGG16, into the FL framework to enhance its predictive capabilities.

Addressing the challenge of data scarcity in some parking lots, Qu et al. [37] propose the Adaptation and Learning to Learn (ALL) approach, combining FL with RL and meta-learning techniques. Their evaluations with data from multiple parking lots demonstrate significant improvements in model variance reduction, training speed, and forecasting performance compared to baseline methods. The study demonstrates that the integration of these advanced ML techniques can effectively address small-sample issues in parking space estimation, providing a robust and scalable solution for smart parking systems. However, challenges, such as communication overhead and model complexity, need to be addressed to fully leverage the potential of FL in the estimation of parking spaces.

While these studies highlight the potential of FL in maintaining data privacy and achieving accurate parking space predictions, they also reveal several challenges. These include scalability and real-world implementation issues, the need for refined incentive mechanisms, optimization of communication protocols to reduce latency and overhead, and the requirement for substantial computational resources for model updates. Moreover, the FL process may experience instability in early training stages, and there is a need to address communication overhead and model complexity to fully leverage the potential of FL in parking space estimation.

Table 3 summarizes the FL applications for the estimation of parking spaces and the research gaps in each method.

Traffic Target Recognition

Traffic target recognition, encompassing object detection, vehicle detection, near-crash detection (NCD), and traffic sign recognition, is a crucial component of ITSs, as described in Figure 4. These tasks are essential to improve road safety, improve traffic management, and enable autonomous driving capabilities. Traditionally, DL methods have shown remarkable success in these areas, leveraging large datasets to train accurate models for various recognition tasks [62], [63], [64].

However, the centralized nature of traditional DL approaches poses significant challenges. These include privacy concerns related to the collection and storage of large amounts of sensitive traffic data, the need for substantial computational resources, and difficulties in adapting to local traffic conditions and regulations that may vary among different regions. Moreover, the dynamic nature of traffic environments requires models that can continuously learn and adapt to new scenarios without compromising historical knowledge. FL has emerged as a promising solution to

Table 3. A summary of FL applications for parking space estimation.

Reference	Model Used	Methodology	Advantages	Research Gaps
[19]	LSTM	FL with parked vehicle-assisted edge computing	Improved data privacy, efficient resource utilization, enhanced computational capacity	Scalability challenges, incentive mechanism refinement, communication overhead
[59]	DRL + optimization	Hierarchical trajectory planning with FL for narrow space automated parking	Improved planning time, maintained trajectory accuracy, decentralized training	Instability in early training stages, computational resource requirements
[36]	ANN, CNN, LSTM	Incorporates dynamic factors, such as parking capacity, occupancy, and temporal variations (holidays, weekends) in FL	Data privacy maintenance, high accuracy in FL setup	Communication overhead, computational resource needs
[37]	ALL (A3C + FL)	Integrated approach for small-sample parking occupancy prediction	Improved model variance reduction, training speed, forecasting performance	Communication overhead, model complexity
[60]	ResNet, DenseNet, EfficientNet, VGG16	FL with spatial and temporal data for parking slot prediction	Enhanced privacy, high predictive accuracy, real-time adaptability	Computational resource requirements

address these limitations while maintaining the benefits of DL for traffic target recognition. FL enables collaborative model training across multiple data sources or edge devices without the need to centralize raw data, thus preserving privacy and allowing for more efficient use of distributed computational resources.

Several researchers have proposed FL models to improve various aspects of traffic target recognition. Jallepalli et al. [38] and Dai et al. [39] focus on improving object detection in autonomous driving systems while preserving data privacy. Their approaches train models locally on vehicles, sharing only parameters with a central server, demonstrating performance comparable to or superior to traditional DL models. These studies highlight FL's potential for rapid optimization and robust performance in critical autonomous driving scenarios, balancing improved object detection with data privacy.

To address the challenge of continuous learning on streaming data, Dai et al. [39] propose the federated learning across autonomous vehicles (FLAME) framework, which performs online FL with continuous streams of data from autonomous vehicles within a simulator. This approach ensures both a dynamic learning process and the generation of new labeled datasets under practical reproducible conditions. Similarly, Chi et al. [41] propose a unified semi-supervised and FL approach that offers cost-efficient and practical training of DL object detection models for autonomous driving, utilizing both labeled and unlabeled data.

Expanding on the concept of collaborative learning, Chi et al. [40] combine FL with cooperative perception in autonomous driving. Their multilayered clustering methodology partitions connected autonomous vehicles (CAVs) and RSUs into local clusters, enabling the sharing of processed perception features among agents. This hybrid method aims to create a more accurate and comprehensive global model that incorporates data captured by both CAVs and RSUs, leading to improved object detection performance.

Addressing security concerns, Mohammed et al. [42] propose a system for secure pedestrian and vehicle detection utilizing homomorphic FL within a distributed fog and cloud network. Their approach demonstrates significant improvements in security accuracy and resource efficiency while also reducing processing time for vehicle and pedestrian detection.

Djenouri et al. [43] introduce a knowledge-guided DL (KGDL) approach (KGDL-NCD) that utilizes an ensemble learning method to build general DL models for crash detection. Their secure FL system groups users into clusters and uses two aggregation methods to form global and customized local models, showing impressive performance on established NCD benchmarks. Xie et al. [61] address the need for privacy-preserving and energy-efficient traffic

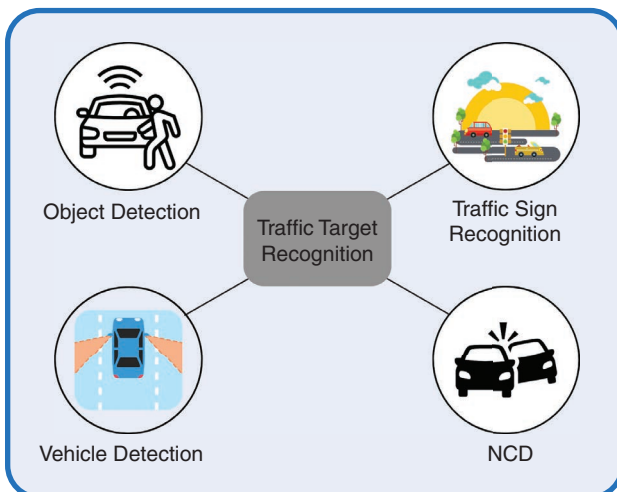


FIG 4 The traffic target recognition.

sign recognition in autonomous vehicles. They propose a novel FL framework integrated with spike neural networks (SNNs) to enhance recognition accuracy while minimizing computational and energy costs.

While these studies demonstrate the potential of FL in various aspects of traffic target recognition, challenges remain. These include the need for more generalized models, integration with real-time traffic data, handling heterogeneous data sources, scalability issues, and reducing communication overhead due to frequent model updates. Table 4 summarizes the use of FL for traffic target recognition, highlighting the diverse methodologies, advantages, and research gaps across different applications.

Intrusion Detection

Intrusion detection systems (IDSs) are crucial for vehicular networks, ensuring the safe and efficient operation of connected transportation systems by detecting and responding to potential cyberthreats and unauthorized access attempts in real time. As vehicles and their systems become more connected and complex, they are increasingly susceptible to cyberattacks that could impact vehicle safety, data privacy, and overall transportation infrastructure security.

To address these challenges, several researchers have proposed innovative approaches using FL for intrusion detection in vehicular networks. Bhavsar et al. [44], Sebastian et al. [65], and Xing et al. [66] have developed systems where edge devices in vehicular networks perform local learning tasks, sharing only model updates or parameters with a central

server. This approach improves data privacy and leverages distributed learning resources without exposing raw data.

Vinita et al. [45] design a framework called federated learning entrusted misbehavior detection system (FLEMDS) that combines FL with software-defined vehicular fog computing networks to detect misbehavior, specifically Sybil attacks. Their system employs a three-level model weight aggregation process and uses fuzzy logic to select vehicles for local FL training, improving attack detection accuracy while minimizing latency.

Campos et al. [46] and Chen et al. [67] have focused on collaborative learning methods for detecting misbehavior in ITSs. Their approaches allow vehicles to train local models on their data and update a global model without sharing data with a central entity, addressing privacy concerns and latency issues associated with centralized ML systems.

These studies have demonstrated several advantages of using FL for intrusion detection in vehicular networks, including enhanced data privacy, improved model accuracy through aggregated learning, and better handling of imbalanced datasets. Xing et al. [66] have further enhanced their approaches by integrating additional techniques, such as memory-augmented autoencoders to improve anomaly detection capabilities.

However, challenges remain in this field, mirroring those seen in other applications of FL in ITSs. Research gaps identified across these studies echo the broader challenges of FL in ITSs, including the need for improved model accuracy at central servers, efficient management of communication

Table 4. A summary of FL applications for traffic target recognition.

Reference	Use Case	Methodology	Advantages	Research Gaps
[38]	Object detection	Decentralized training of object detection models on each vehicle's dataset	Minimizes communication overhead and maintains privacy	Need for consistent image label formats and model configurations across clients
[39]	Object detection	FLAME framework for online FL with continuous data streams in a simulator	Dynamic learning process, better generalization and performance	Communication overhead and synchronization challenges in dynamic real-time environments
[41]	Object detection	Local model training using both labeled and unlabeled data	Improved object detection capabilities	High latency and overhead associated with model updates
[40]	Object detection and vehicle detection	Multilayered clustering method for sharing perception features among CAVs and RSUs using V2X communication	Comprehensive global model	Complexity of managing multiple clusters and ensuring robust V2X communication
[42]	Object detection and vehicle detection	Integrates homomorphic encryption with FL to ensure secure and efficient data processing across distributed fog and cloud networks	High security accuracy, reduced resource leakage, faster processing time	Potential scalability issues, managing resource allocation in fog and cloud environments
[43]	NCD	KGDL-NCD using ensemble learning and k -nearest neighbors for model selection	High AUC metric (0.95), effective NCD, customized local models	Need for more generalized models, integration with real-time traffic data
[61]	Traffic sign recognition	FL with SNNs and NRFE	Privacy preservation, low power consumption, fast inference, noise robustness	Unable to handle heterogeneous data sources

overhead between edge devices and central servers, vulnerability to adversarial attacks, and the need for efficient handling of nonindependent identically distributed data distributions. These challenges are particularly pronounced in IDSs, due to the critical nature of security in vehicular networks.

As with other ITS applications of FL, such as traffic flow prediction and autonomous driving, the intrusion detection domain is grappling with the heterogeneity of data across different vehicles and locations. This data heterogeneity can lead to model inconsistencies and reduced overall performance. Additionally, the resource constraints of edge devices in vehicles, a common theme across FL applications in ITSs, pose significant challenges for implementing complex intrusion detection models.

Open Challenges

This section presents various open challenges that exist in enabling FL over ITSs.

Data Heterogeneity and Trustworthiness

Data heterogeneity refers to the differences in the structure, format, and distribution of data across different ITS environments. This heterogeneity can significantly affect the deployment and effectiveness of the collaborative model training process. One of the primary challenges posed by heterogeneity in the data is the potential disparity in the quality and quantity of data [29], [32]. This can lead to unequal participation in and contribution to the FL process, ultimately impacting the fairness and representativeness of the global model. For instance, the FedSTN framework [29] had to address varying spatiotemporal data characteristics across different traffic scenarios, highlighting the challenge of integrating diverse data sources. Additionally, data heterogeneity can introduce complexities in aggregating and reconciling model updates from diverse sources, potentially leading to convergence issues and suboptimal model performance.

Furthermore, disparate data characteristics, such as varying feature distributions, data types, and domain-specific attributes, can pose compatibility challenges during the aggregation and fusion of local model updates. This can hinder the seamless integration of diverse data sources into the global model, affecting the model's overall accuracy and generalization capabilities across heterogeneous ITS environments.

In addition to heterogeneity, the trustworthiness of the data plays a crucial role in the successful development of FL [65], [66]. Since FL relies on multiple parties to collaborate in building a shared global model without sharing their local data, the trustworthiness of the data becomes paramount. If the data from participating devices and systems are not reliable or trustworthy, they can lead to inaccurate model convergence and compromise the overall effectiveness of the FL method in ITSs.

Unreliable data can introduce biases and distortions in the shared model, leading to suboptimal decision mak-

ing and performance in ITS applications. Furthermore, in the context of FL, where data are collected from various sources, such as traffic cameras, GPS devices, and vehicle sensors, ensuring the trustworthiness and consistency of the data becomes even more challenging [38], [40].

Furthermore, trust issues with the shared model and the local training data across heterogeneous and untrusted edge devices can exacerbate privacy and security concerns. Establishing mechanisms to verify and ensure the trustworthiness of the data from diverse sources is crucial for the successful deployment of FL in ITSs. Only when the data can be trusted can FL achieve its potential to improve the intelligence and efficiency of transportation systems.

Communication and Bandwidth Constraints

Communication and bandwidth constraints pose significant challenges to the development of FL in ITSs. The dynamic and distributed nature of transportation networks, coupled with the high volume and velocity of real-time traffic data, exacerbates these challenges. Studies by Liu et al. [27] and Sepasgozar et al. [28] highlight the substantial communication overhead resulting from frequent model updates in applications such as real-time traffic flow prediction. This issue is particularly pronounced when dealing with complex DL models, as seen in the work of Qi et al. [31] on their FedAGCN framework.

Transmitting model parameters across the network for each update consumes significant bandwidth, especially when dealing with complex DL models containing millions of parameters. This issue is compounded by the heterogeneous nature of ITS infrastructure, where RSUs, vehicles, and traffic control centers may have varying levels of communication capabilities and bandwidth availability. Such heterogeneity can create bottlenecks in the FL process, with some nodes struggling to maintain the update frequency necessary for effective learning.

In addition, the mobile nature of many ITS components, particularly vehicles, further complicates the situation. Vehicles traversing areas with poor network coverage may experience intermittent connectivity, potentially disrupting the FL process and leading to model inconsistencies or reduced performance. The need for low-latency communication in critical ITS applications, such as collision avoidance systems, adds further stress to the network.

These constraints can result in delays in model synchronization, increased communication overhead, and difficulties in coordinating the learning process among distributed devices. Consequently, the overall efficiency and performance of FL in ITSs may be significantly impacted. Balancing the requirements of timely model updates with the constraints of available bandwidth presents a delicate optimization problem that requires innovative solutions to ensure the effective implementation of FL in ITS environments.

Resource-Constrained Edge Devices

Resource limitations of edge devices in ITS environments pose significant challenges to the effective implementation of FL. These challenges stem from the inherent constraints of ITS edge devices, such as traffic cameras, vehicle sensors, GPS devices, RSUs, and IoT sensors, which often possess limited computational capabilities, finite energy sources, and restricted storage capacities. The deployment of FL in such resource-constrained environments can significantly affect FL's effectiveness and efficiency [61]. Running sophisticated ML algorithms and participating in iterative model training processes can strain the resources of these devices, leading to several critical deployment challenges.

The mismatch between the computational demands of complex ML models and the available resources can lead to prolonged training times and reduced model performance, potentially compromising the real-time responsiveness crucial in ITS applications [42]. The limited computational power may restrict the complexity of ML models that can be trained on the edge, affecting the overall learning and inference capabilities of the federated model. Moreover, the energy-intensive nature of FL, involving frequent local computations and model updates, presents a significant challenge in balancing model accuracy with energy consumption to ensure the longevity and sustainability of ITS infrastructure. This is particularly crucial in scenarios where continuous operation is essential, such as real-time traffic monitoring and management, as FL algorithms require frequent communication and computation rounds, which can drain the limited battery life of edge devices.

The storage limitations of edge devices further constrain the quantity and quality of data that can be utilized for local training, potentially impacting the overall performance of the federated model [65]. Constrained storage capacity can limit the ability of edge devices to store and process large model updates and training data, potentially causing inefficiencies in the learning process. These issues are compounded by the stringent real-time processing requirements of many ITS applications, where the iterative nature of FL may introduce unacceptable latency [59].

The heterogeneity of the ITS ecosystem, consisting of diverse edge devices with varying capabilities, further complicates the design of FL systems that must accommodate the weakest devices without significantly compromising overall performance. At the core of these challenges lies a fundamental tension between the desire for increasingly sophisticated ML models to address complex ITS challenges and the resource constraints of edge devices.

Future Directions

The implementation of FL in ITSs faces several significant challenges, primarily stemming from data heterogeneity

and trustworthiness, communication and bandwidth constraints, and resource-constrained edge devices. Below are potential future directions and solutions that researchers and practitioners may explore to address these challenges and advance the field:

- *Robust and secure aggregation methods:* Developing advanced aggregation techniques that are resistant to adversarial attacks and can handle data discrepancies is essential for ensuring robustness in the collective model. This is particularly important in ITS environments where security and reliability are paramount.
- *Improving model personalization:* This involves exploring methods that can adapt the federated model to local contexts better, such as metalearning or multitask learning, to improve model performance for individual users while maintaining the advantages of FL. This personalization could lead to more accurate and relevant models for specific ITS scenarios or geographical areas, addressing the challenge of data heterogeneity.
- *Federated transfer learning:* Incorporating transfer learning principles into FL frameworks could help address data heterogeneity by leveraging pretrained models and adapting them to specific ITS tasks. This approach could reduce the overall training time and resource requirements while improving model performance across diverse data distributions.
- *Communication-efficient techniques:* This involves developing efficient compression techniques for model updates and exploring communication-efficient FL algorithms to reduce the overall bandwidth requirements of the FL process.
- *Adaptive resource allocation:* This relates to designing adaptive FL frameworks that can dynamically adjust to varying network conditions and bandwidth availability, intelligently allocating tasks and prioritizing critical operations across the network.
- *Lightweight and efficient FL algorithms:* Work here centers on developing FL algorithms specifically tailored for resource-constrained environments, focusing on creating lightweight versions of existing FL algorithms or entirely new approaches that minimize computational and memory requirements.
- *Optimization for heterogeneous networks:* This concerns developing algorithms that are adept at handling the highly heterogeneous and dynamic nature of vehicular networks regarding data distribution, computational power, and network connectivity. This optimization could address challenges across all three main problem areas.
- *Standardization and benchmarking:* Developing standardized benchmarks, evaluation metrics, and comprehensive evaluation frameworks specifically for FL in ITSs would facilitate comparisons among different approaches and accelerate progress in addressing all identified challenges. This includes

establishing standards for FL implementations, communication protocols, and data formats to ensure interoperability in ITSs.

The successful implementation of these approaches could lead to more efficient, scalable, and robust FL systems for ITSs, ultimately enhancing the intelligence and responsiveness of transportation networks while addressing the challenges of data heterogeneity, communication constraints, and resource limitations of edge devices.

Conclusion

The integration of FL in ITSs marks a significant advancement in managing modern transportation networks, offering enhanced data privacy, scalability, and resource efficiency. However, challenges such as data heterogeneity, communication limitations, and resource constraints on edge devices require innovative solutions. This article has highlighted the need for customized FL solutions in ITSs and provided a road map for future research, emphasizing interdisciplinary approaches to address these multifaceted challenges. Future efforts should focus on developing robust FL frameworks that accommodate the dynamic nature of transportation systems, advancing communication technologies, optimizing resource allocation, and ensuring data trustworthiness. Overcoming these challenges will pave the way for more responsive and efficient ITS solutions, capable of leveraging distributed edge devices while respecting their limitations. As the field progresses, the continued refinement of FL techniques in ITSs will play a crucial role in shaping intelligent and adaptive transportation networks, ultimately contributing to improved urban mobility, enhanced safety, and sustainable transportation management in ever-evolving urban environments.

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References

- [1] “Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Definitions.” ETSI. Accessed: Dec. 15, 2023. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/102600_102699/102638/01.01.01_60/tr_102638v010101p.pdf
- [2] X. Wang, W. Liu, H. Lin, J. Hu, K. Kaur, and M. S. Hossain, “AI-empowered trajectory anomaly detection for intelligent transportation systems: A hierarchical federated learning approach,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4631–4640, Apr. 2023, doi: [10.1109/TITS.2022.3209903](https://doi.org/10.1109/TITS.2022.3209903).
- [3] D. M. Manias and A. Shami, “Making a case for federated learning in the internet of vehicles and intelligent transportation systems,” *IEEE Netw.*, vol. 35, no. 3, pp. 88–94, May/Jun. 2021, doi: [10.1109/MNET.011.2000552](https://doi.org/10.1109/MNET.011.2000552).
- [4] A. Haydari and Y. Y. Ilmaz, “Deep reinforcement learning for intelligent transportation systems: A survey,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 1, pp. 11–32, Jan. 2022, doi: [10.1109/TITS.2020.3008612](https://doi.org/10.1109/TITS.2020.3008612).
- [5] F.-Y. Wang et al., “Transportation 5.0: The DAO to safe, secure, and sustainable intelligent transportation systems,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 10,262–10,278, Oct. 2023, doi: [10.1109/TITS.2023.3305380](https://doi.org/10.1109/TITS.2023.3305380).
- [6] K. Bonawitz et al., “Towards federated learning at scale: System design,” *Proc. Mach. Learn. Syst.*, vol. 1, pp. 374–388, Feb. 2019.
- [7] Y. Lv, Y. Duan, W. Kang, Z. Li, and F.-Y. Wang, “Traffic flow prediction with big data: A deep learning approach,” *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 2, pp. 865–875, Apr. 2015, doi: [10.1109/TITS.2014.2345663](https://doi.org/10.1109/TITS.2014.2345663).
- [8] R. Zhang, H. Wang, B. Li, X. Cheng, and L. Yang, “A survey on federated learning in intelligent transportation systems,” 2024, *arXiv:2403.07444*.
- [9] Q. Yang, Y. Liu, T. Chen, and Y. Tong, “Federated machine learning: Concept and applications,” *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 1–19, 2019, doi: [10.1145/3298981](https://doi.org/10.1145/3298981).
- [10] L. U. Khan, W. Saad, Z. Han, E. Hossain, and C. S. Hong, “Federated learning for internet of things: Recent advances, taxonomy, and open challenges,” *IEEE Commun. Surveys Tut.*, vol. 23, no. 3, pp. 1759–1799, 3rd Quart. 2021, doi: [10.1109/COMST.2021.3090430](https://doi.org/10.1109/COMST.2021.3090430).
- [11] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. 20th Int. Conf. Artif. Intell. Statist. (AISTATS)*, Ft. Lauderdale, FL, USA, Apr. 2017, pp. 1273–1282.
- [12] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” 2018, *arXiv:1812.06127*.
- [13] P. Kairouz et al., “Advances and open problems in federated learning,” 2019, *arXiv:1912.04977*.
- [14] W. Y. B. Lim et al., “Federated learning in mobile edge networks: A comprehensive survey,” *IEEE Commun. Surveys Tut.*, vol. 22, no. 3, pp. 2051–2063, Apr. 2020, doi: [10.1109/COMST.2020.2986024](https://doi.org/10.1109/COMST.2020.2986024).
- [15] T. Luo, J. Huang, S. S. Kanhere, J. Zhang, and S. K. Das, “Improving IoT data quality in mobile crowd sensing: A cross validation approach,” *IEEE Internet Things J.*, vol. 6, no. 3, pp. 5651–5664, Jun. 2019, doi: [10.1109/JIOT.2019.2904704](https://doi.org/10.1109/JIOT.2019.2904704).
- [16] S. Liu, Z. Zheng, F. Wu, S. Tang, and G. Chen, “Context-aware data quality estimation in mobile crowdsensing,” in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, 2017, pp. 1–9.
- [17] C. Zhao, S. Yang, P. Yan, Q. Yang, X. Yang, and J. McCann, “Data quality guarantee for credible caching device selection in mobile crowdsensing systems,” *IEEE Wireless Communications*, vol. 25, no. 3, pp. 58–64, Jun. 2018, doi: [10.1109/MWC.2018.1700299](https://doi.org/10.1109/MWC.2018.1700299).
- [18] Z. Du, C. Wu, T. Yoshinaga, K.-L. A. Yau, Y. Ji, and J. Li, “Federated learning for vehicular internet of things: Recent advances and open issues,” *IEEE Open J. Comput. Society*, vol. 1, pp. 45–61, 2020, doi: [10.1109/OJCS.2020.2992630](https://doi.org/10.1109/OJCS.2020.2992630).
- [19] X. Huang, P. Li, R. Yu, Y. Wu, K. Xie, and S. Xie, “FedParking: A federated learning based parking space estimation with parked vehicle assisted edge computing,” *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 9355–9368, Sep. 2021, doi: [10.1109/TVT.2021.3098170](https://doi.org/10.1109/TVT.2021.3098170).
- [20] F. Mao, Z. Li, and L. Li, “A comparison of deep reinforcement learning models for isolated traffic signal control,” *IEEE Intell. Transp. Syst. Mag.*, vol. 15, no. 1, pp. 160–180, Jan./Feb. 2022, doi: [10.1109/ITS.2022.3144797](https://doi.org/10.1109/ITS.2022.3144797).
- [21] B. R. Kiran et al., “Deep reinforcement learning for autonomous driving: A survey,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, Jun. 2022, doi: [10.1109/TITS.2021.3054625](https://doi.org/10.1109/TITS.2021.3054625).
- [22] S. Zhang et al., “Federated learning in intelligent transportation systems: Recent applications and open problems,” *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 5, May 2023, doi: [10.1109/TITS.2023.3524962](https://doi.org/10.1109/TITS.2023.3524962).
- [23] B. Ghimire and D. B. Rawat, “Secure, privacy preserving, and verifiable federated learning using blockchain for internet of vehicles,” *IEEE Consum. Electron. Mag.*, vol. 11, no. 6, pp. 67–74, Nov. 2022, doi: [10.1109/MCE.2021.3097705](https://doi.org/10.1109/MCE.2021.3097705).
- [24] A. Brecko, E. Kajati, J. Koziorek, and I. Zolotova, “Federated learning for edge computing: A survey,” *Appl. Sci.*, vol. 12, no. 18, 2022, Art. no. 9124, doi: [10.3390/app12189124](https://doi.org/10.3390/app12189124).
- [25] Y. Qi, M. S. Hossain, J. Nie, and X. Li, “Privacy-preserving blockchain-based federated learning for traffic flow prediction,” *Future Gener. Comput. Syst.*, vol. 117, pp. 328–337, Apr. 2021, doi: [10.1016/j.future.2020.12.003](https://doi.org/10.1016/j.future.2020.12.003).
- [26] L. Hu et al., “A dynamic pyramid tilling method for traffic data stream based on Flink,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6679–6688, Jul. 2022, doi: [10.1109/TITS.2021.3060576](https://doi.org/10.1109/TITS.2021.3060576).

- [27] Y. Liu, J. Q. James, J. Kang, D. Niyato, and S. Zhang, "Privacy-preserving traffic flow prediction: A federated learning approach," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7751–7763, Aug. 2020, doi: [10.1109/JIOT.2020.2991401](#).
- [28] S. S. Sepasgozar and S. Pierre, "Fed-NTP: A federated learning algorithm for network traffic prediction in Vanet," *IEEE Access*, vol. 10, pp. 119,607–119,616, 2022, doi: [10.1109/ACCESS.2022.3221970](#).
- [29] X. Yuan et al., "FedSTN: Graph representation driven federated learning for edge computing enabled urban traffic flow prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 8, pp. 8738–8748, Aug. 2023, doi: [10.1109/TITS.2022.3157056](#).
- [30] S. Samarakoon, M. Bennis, W. Saad, and M. Debbah, "Federated learning for ultra-reliable low-latency V2V communications," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Piscataway, NJ, USA: IEEE Press, 2018, pp. 1–7.
- [31] T. Qi, L. Chen, G. Li, Y. Li, and C. Wang, "FedAGCN: A traffic flow prediction framework based on federated learning and Asynchronous Graph Convolutional Network," *Appl. Soft Comput.*, vol. 138, May 2023, Art. no. 110175, doi: [10.1016/j.asoc.2023.110175](#).
- [32] M. Xia, D. Jin, and J. Chen, "Short-term traffic flow prediction based on graph convolutional networks and federated learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 1191–1203, Jan. 2023, doi: [10.1109/TITS.2022.3179391](#).
- [33] C. Wang, L. Ma, R. Li, T. S. Durrani, and H. Zhang, "Exploring trajectory prediction through machine learning methods," *IEEE Access*, vol. 7, pp. 101,441–101,452, 2019, doi: [10.1109/ACCESS.2019.2929450](#).
- [34] X. Zhou, R. Ke, Z. Cui, Q. Liu, and W. Qian, "STFL: Spatio-temporal federated learning for vehicle trajectory prediction," in *Proc. IEEE 2nd Int. Conf. Digital Twins Parallel Intell. (DTPDI)*, Piscataway, NJ, USA: IEEE Press, 2022, pp. 1–6, doi: [10.1109/DTPDI55838.2022.9998967](#).
- [35] C. Koetsier, J. Fiosina, J. N. Gremmel, J. P. Müller, D. M. Woitschlager, and M. Sester, "Detection of anomalous vehicle trajectories using federated learning," *ISPRS Open J. Photogrammetry Remote Sens.*, vol. 4, Apr. 2022, Art. no. 100015, doi: [10.1016/j.ojphoto.2022.100015](#).
- [36] S. Senapati, S. Mallik, V. S. Negi, and S. Chinara, "Comparison of different models for federated learning based parking space estimation," in *Proc. 14th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, Piscataway, NJ, USA: IEEE Press, 2023, pp. 1–6, doi: [10.1109/ICCCNT56998.2023.10506879](#).
- [37] H. Qu, S. Liu, J. Li, Y. Zhou, and R. Liu, "Adaptation and learning to learn (ALL): An integrated approach for small-sample parking occupancy prediction," *Mathematics*, vol. 10, no. 12, 2022, Art. no. 2059, doi: [10.3390/math10122059](#).
- [38] D. Jallepalli, N. C. Ravikumar, P. V. Badarinarath, S. Uchil, and M. A. Suresh, "Federated learning for object detection in autonomous vehicles," in *Proc. IEEE 7th Int. Conf. Big Data Comput. Service Appl. (BigDataService)*, Piscataway, NJ, USA: IEEE Press, 2021, pp. 107–114, doi: [10.1109/BigDataService52369.2021.00018](#).
- [39] S. Dai, S. M. Iftikharul Alam, R. Balakrishnan, K. Lee, S. Banerjee, and N. Himayat, "Online federated learning based object detection across autonomous vehicles in a virtual world," in *Proc. IEEE 20th Consum. Commun. Netw. Conf. (CCNC)*, Piscataway, NJ, USA: IEEE Press, 2023, pp. 919–920, doi: [10.1109/CCNC51644.2023.10060782](#).
- [40] F. Chi, Y. Wang, P. Nasiopoulos, and V. C. M. Leung, "Federated cooperative 3D object detection for autonomous driving," in *Proc. IEEE 33rd Int. Workshop Mach. Learn. Signal Process. (MLSP)*, Piscataway, NJ, USA: IEEE Press, 2023, pp. 1–6, doi: [10.1109/MLSP55844.2023.10285959](#).
- [41] F. Chi, Y. Wang, P. Nasiopoulos, V. C. M. Leung, and M. T. Pourazad, "Federated semi-supervised learning for object detection in autonomous driving," in *Proc. IEEE international conference on acoustics, speech and signal processing (ICASSP)*, Piscataway, NJ, USA: IEEE Press, 2023, pp. 1–5.
- [42] M. A. Mohammed et al., "Homomorphic federated learning schemes enabled pedestrian and vehicle detection system," *Internet Things*, vol. 25, Oct. 2023, Art. no. 100903, doi: [10.1016/j.iot.2023.100903](#).
- [43] Y. Djenouri, A. N. Belbachir, T. Michalak, A. Belhadi, and G. Srivastava, "Enhancing smart road safety with federated learning for Near Crash Detection to advance the development of the internet of vehicles," *Eng. Appl. Artif. Intell.*, vol. 133, Jul. 2024, Art. no. 108350, doi: [10.1016/j.engappai.2024.108350](#).
- [44] M. Bhavsar, Y. Bekele, K. Roy, J. Kelly, and D. Limbrick, "FL-IDS: Federated learning-based intrusion detection system using edge devices for transportation IoT," *IEEE Access*, vol. 12, pp. 52,215–52,226, 2024, doi: [10.1109/ACCESS.2024.3586631](#).
- [45] L. J. Vinita and V. Vetrivel, "Federated learning-based misbehaviour detection on an emergency message dissemination scenario for the 6G-enabled internet of vehicles," *Ad Hoc Networks*, vol. 144, May 2023, Art. no. 103153, doi: [10.1016/j.adhoc.2023.103153](#).
- [46] E. M. Campos, J. L. Hernandez-Ramos, A. González Vidal, G. Baldini, and A. Skarmeta, "Misbehavior detection in intelligent transportation systems based on federated learning," *Internet Things*, vol. 25, Apr. 2024, Art. no. 101127, doi: [10.1016/j.iot.2024.101127](#).
- [47] M. Xu, Y. Di, H. Ding, Z. Zhu, X. Chen, and H. Yang, "AGNP: Network-wide short-term probabilistic traffic speed prediction and imputation," *Commun. Transp. Res.*, vol. 3, Dec. 2023, Art. no. 100099, doi: [10.1016/j.commtr.2023.100099](#).
- [48] Y. Liu, C. Lyu, Y. Zhang, Z. Liu, W. Yu, and X. Qu, "DeepTSP: Deep traffic state prediction model based on large-scale empirical data," *Commun. Transp. Res.*, vol. 1, Dec. 2021, Art. no. 100012, doi: [10.1016/j.commtr.2021.100012](#).
- [49] Y. Chen, B. Ivanovic, and M. Pavone, "ScePT: Scene-consistent, policy-based trajectory predictions for planning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 17,103–17,112.
- [50] C. Wang, X. Chen, J. Wang, and H. Wang, "ATPFL: Automatic trajectory prediction model design under federated learning framework," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, pp. 6563–6572, 2022, doi: [10.1109/CVPR52688.2022.00645](#).
- [51] W. Zhu, J. Wu, T. Fu, J. Wang, J. Zhang, and Q. Shangguan, "Dynamic prediction of traffic incident duration on urban expressways: A deep learning approach based on LSTM and MLP," *J. Intell. Connected Vehicles*, vol. 4, pp. 80–91, 2021, doi: [10.1108/JICV-05-2021-0004](#).
- [52] C. N. Babu, P. Sure, and C. M. Bhumra, "Sparse Bayesian learning assisted approaches for road network traffic state estimation," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1733–1741, Mar. 2021, doi: [10.1109/TITS.2020.2971031](#).
- [53] H. T. Vu and C.-C. Huang, "A multi-task convolutional neural network with spatial transform for parking space detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Beijing, China, pp. 1762–1766, 2017.
- [54] K. Gkolias and E. I. Vlahogianni, "Convolutional neural networks for on-street parking space detection in urban networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 12, pp. 4318–4327, Dec. 2019, doi: [10.1109/TITS.2018.2882459](#).
- [55] R. Kaur, R. K. Roul, and S. Batra, "A hybrid deep learning CNN-ELM approach for parking space detection in Smart Cities," *Neural Comput. Appl.*, vol. 35, no. 18, pp. 13,665–13,685, 2023, doi: [10.1007/s00521-023-08426-y](#).
- [56] T. Anagnostopoulos, P. Fedchenkov, N. Tsotsolas, K. Ntalianis, A. Zaslavsky, and I. Salmon, "Distributed modeling of smart parking system using LSTM with stochastic periodic predictions," *Neural Comput. Appl.*, vol. 32, no. 14, pp. 10,783–10,796, 2020, doi: [10.1007/s00521-019-04613-y](#).
- [57] J. Fan, Q. Hu, Y. Xu, and Z. Tang, "Predicting vacant parking space availability: A long short-term memory approach," *IEEE Intell. Transp. Syst. Mag.*, vol. 14, no. 2, pp. 129–143, Mar./Apr. 2022, doi: [10.1109/MTS.2020.3014131](#).
- [58] R. K. Kasera and T. Acharjee, "Parking slot occupancy prediction using LSTM," *Innovations Syst. Softw. Eng.*, pp. 1–13, Aug. 2022, doi: [10.1007/s11354-022-00481-3](#).
- [59] Z. Yuan, Z. Wang, X. Li, L. Li, and L. Zhang, "Hierarchical trajectory planning for narrow-space automated parking with deep reinforcement learning: A federated learning scheme," *Sensors*, vol. 23, no. 8, 2023, Art. no. 4087, doi: [10.3390/s23084087](#).
- [60] A. Sha, J. John, M. Varundev, S. Surendran, S. Abhishek, and A. T, "Urban parking management through federated learning: A privacy-preserving approach to parking slot prediction," in *Proc 2nd Int. Conf. Automat., Comput. Renewable Syst. (ICACRS)*, Pudukkottai, India, pp. 896–901, 2023, doi: [10.1109/ICACRS58579.2023.10404706](#).
- [61] K. Xie et al., "Efficient federated learning with spike neural networks for traffic sign recognition," *IEEE Trans. Veh. Technol.*, vol. 71, no. 9, pp. 9980–9992, Sep. 2022, doi: [10.1109/TVT.2022.3178808](#).
- [62] Y. Zhu and W. Q. Yan, "Traffic sign recognition based on deep learning," *Multimedia Tools Appl.*, vol. 81, no. 13, pp. 17,779–17,791, 2022, doi: [10.1007/s11042-022-12165-0](#).
- [63] S. Li, J. Lin, G. Li, T. Bai, H. Wang, and Y. Pang, "Vehicle type detection based on deep learning in traffic scene," *Procedia Comput. Sci.*, vol. 131, pp. 564–572, May 2018.
- [64] D. Li, D. Zhao, Y. Chen, and Q. Zhang, "Deepsign: Deep learning based traffic sign recognition," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Rio de Janeiro, Brazil, pp. 1–6, 2018, doi: [10.1109/IJCNN.2018.8489623](#).
- [65] A. Sebastian, "Enhancing intrusion detection in internet of vehicles through federated learning," 2023, *arXiv:2311.13800*.
- [66] L. Xing, K. Wang, H. Wu, H. Ma, and X. Zhang, "FL-MAAE: An intrusion detection method for the internet of vehicles based on federated learning and memory-augmented autoencoder," *Electronics*, vol. 12, no. 10, 2023, Art. no. 2284, doi: [10.3390/electronics12102284](#).
- [67] X. Chen, W. Qiu, L. Chen, Y. Ma, and J. Ma, "Fast and practical intrusion detection system based on federated learning for VANET," *Comput. Secur.*, vol. 142, Jul. 2024, Art. no. 105881, doi: [10.1016/j.cose.2024.105881](#).